AI Agents

Michiel Bontenbal

Sensemakers Amsterdam

15 april 2026

sense
maker
samst
erdam

My name is Michiel

I live in Amsterdam

I work at Hogeschool van Amsterdam

My secret superpower is having white hair.

My goal is to teach everyone about AI.

I have a smile on my face when

I see happy people around me

https://linktr.ee/mpbontenbal

Agenda

sense
maker
samst
erdam

1. Intro
2. From LLM's to Agents
    1. Exercise: get started with a simple framework
3. Some agentic frameworks
    1. Exercise: compare code
4. Agents connecting with tools: MCP and Skills
    1. Start with MCP and Skills
5. Danger Zone: autonomous agents

Clone the repo & install ollama

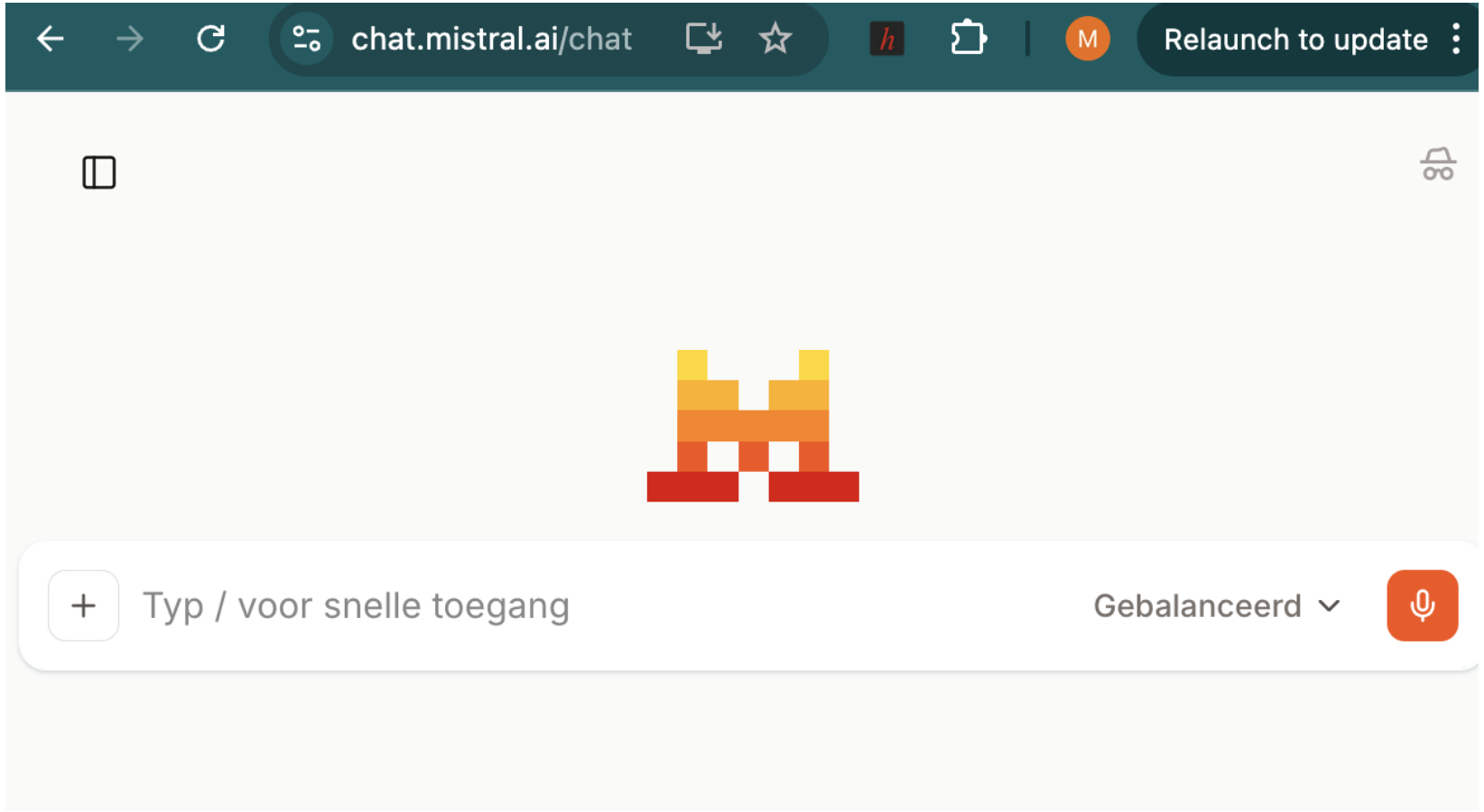
sense
maker
samst
erdam

- https://github.com/MichielBontenbal/agent_lecture
- or: http://tiny.cc/j5w1101

- We will use ollama to serve models.
- Download it from www.ollama.com

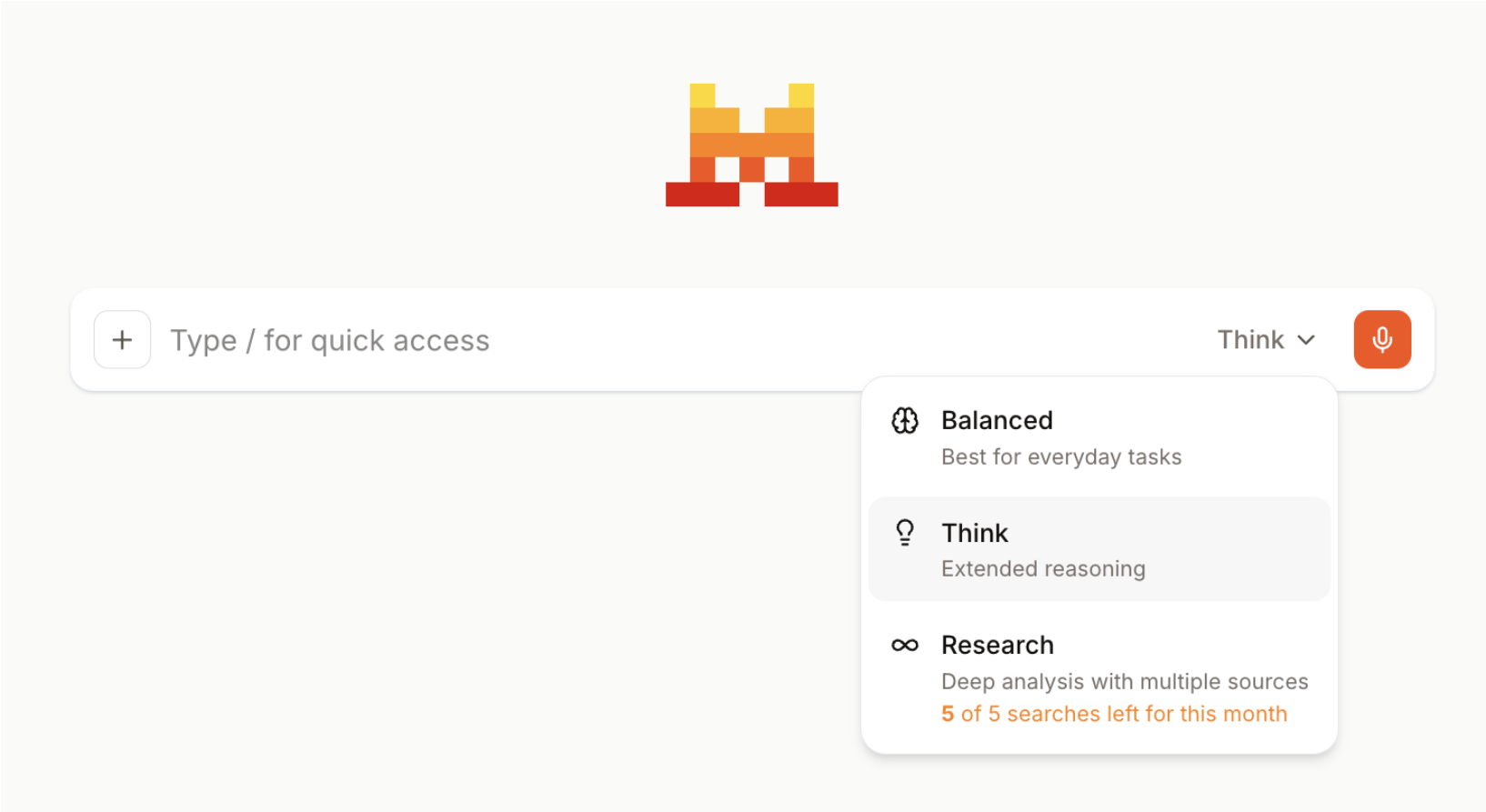
1. From LLM’s to Agents

You know the LLM...



sense
maker
samst
erdam

...and the LLM can reason ('Think') now...



sense
maker
samst
erdam

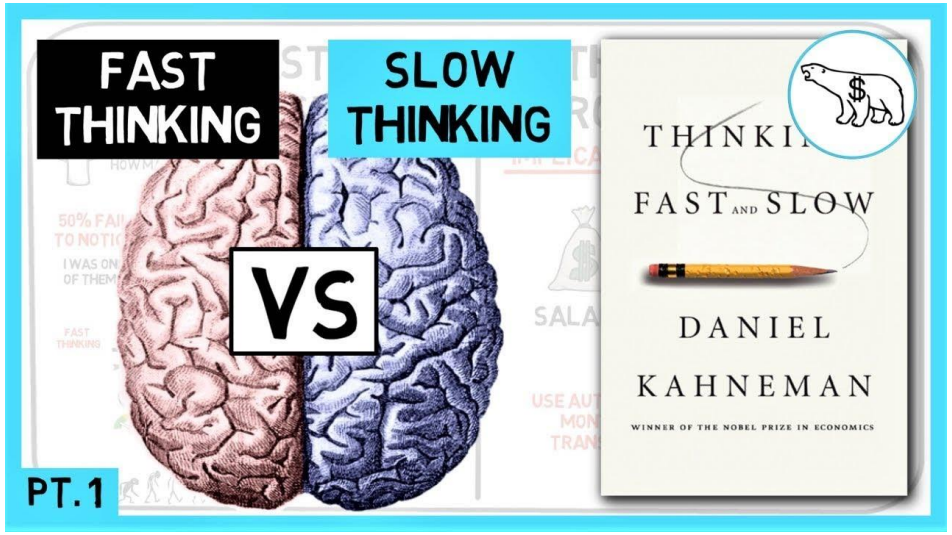
Since ± 2024 LLM's can reason

Fast thinking:

e.g. what is the capital of France?

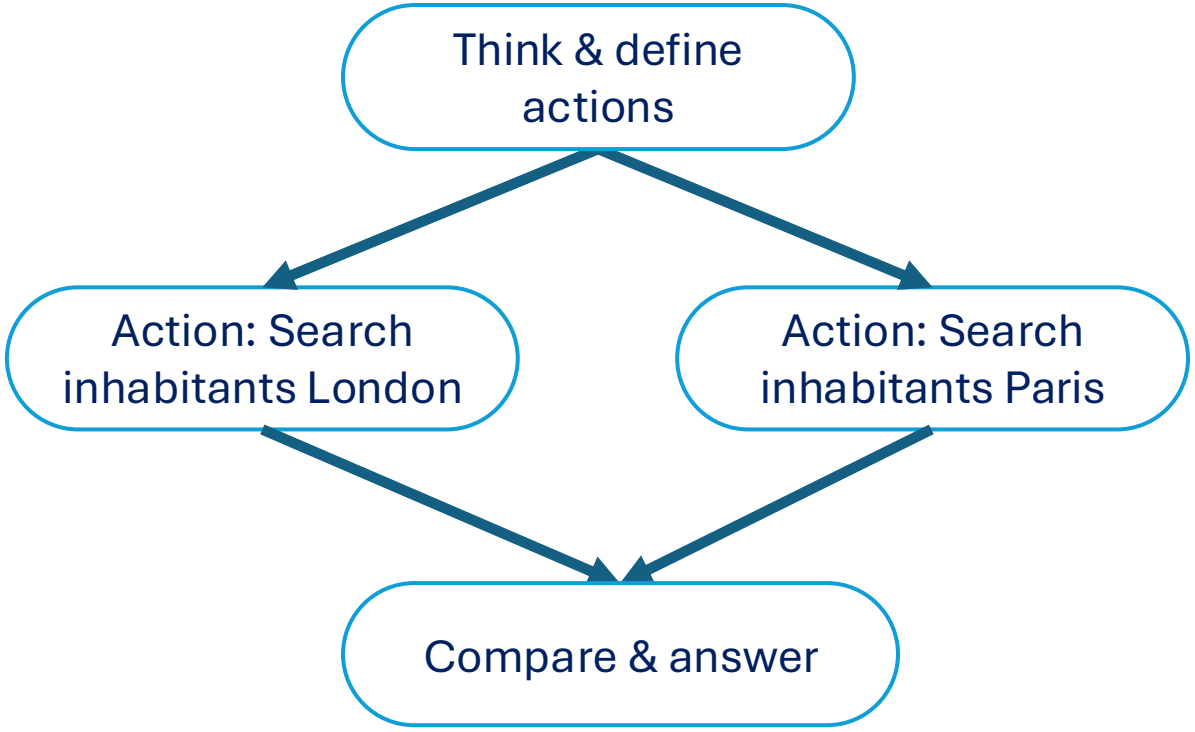
Slow thinking:

e.g. which city has more people, Paris or London?



Question for you: *how* to solve this?

Which city has more inhabitants, Paris or London?



Mistral is not only an LLM, but can use search tool

Thought for 5s

Thinking

The user is asking about population comparison between Paris and London. I need to provide current, accurate data. This is straightforward factual information that I should verify from reliable sources. I'll perform two web searches to get the latest population figures for both cities, using 2026 as the reference year to ensure currency.

Searched web "Paris population 2026"

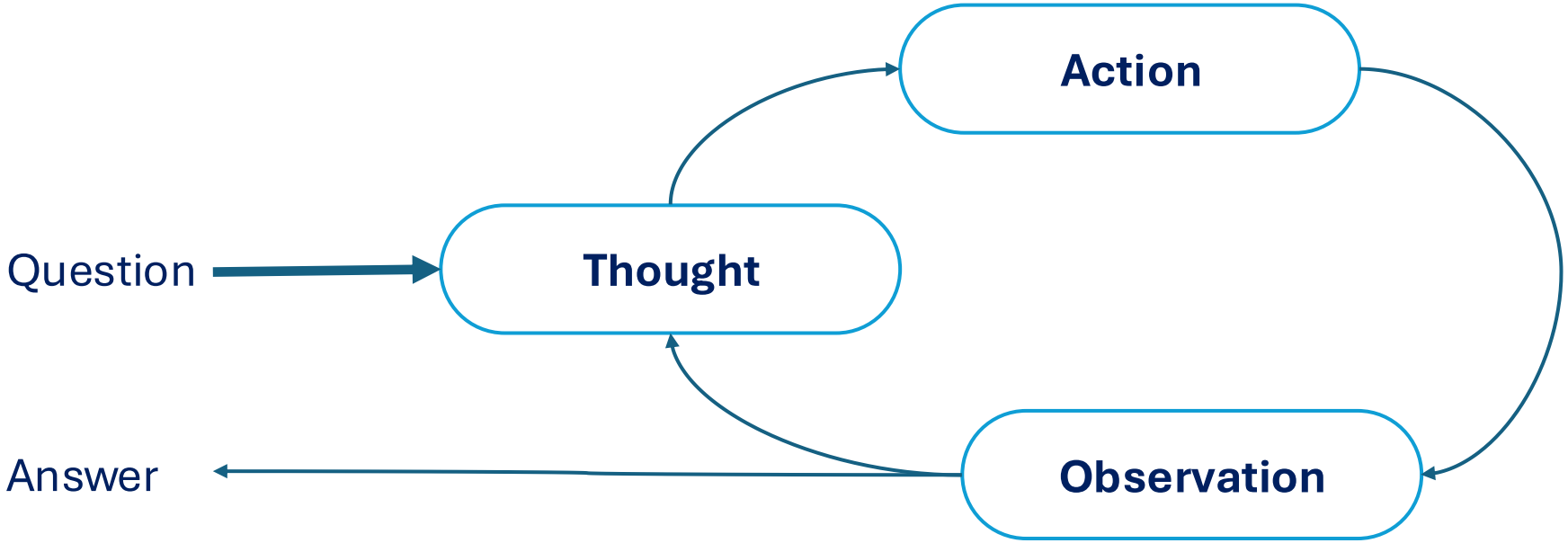
Searched web "London population 2026"

Thinking

I've received the search results for both cities' populations. The data shows Paris metropolitan area at approximately 13.2 million and London at approximately 9.9 million for 2026. Paris has a significantly larger population. I should present this clearly, cite the sources properly, and offer to provide more details if the user wants them.

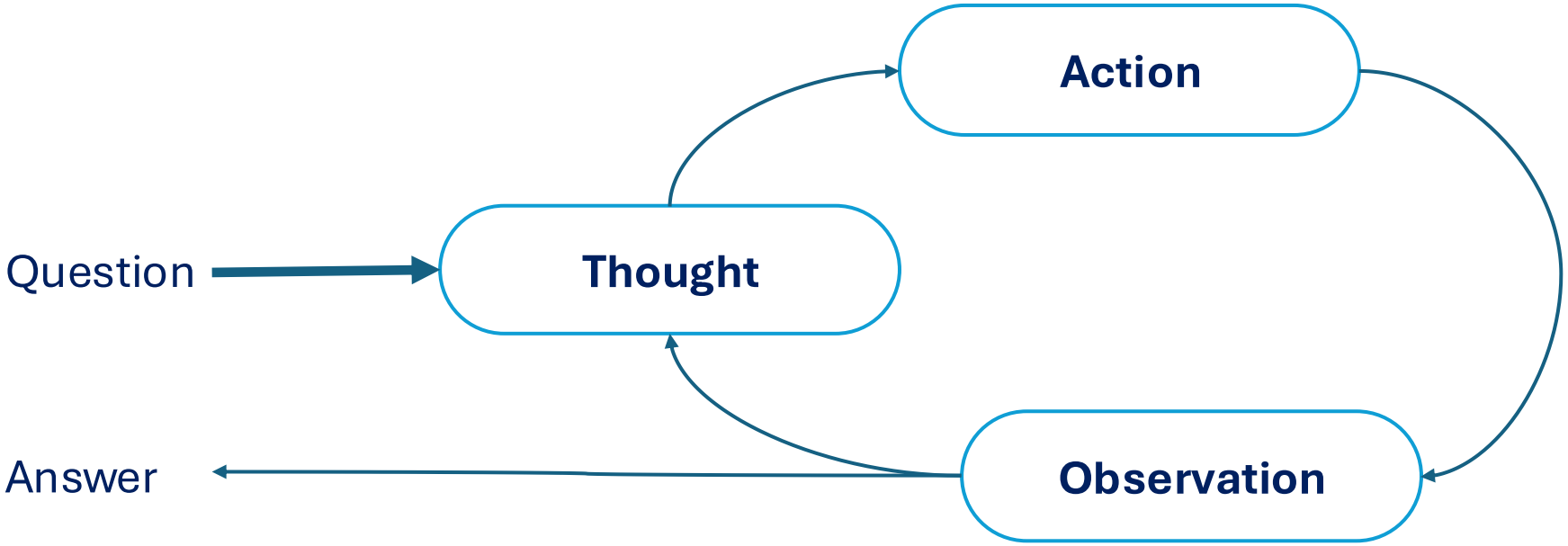
As of 2026, **London has a larger population than Paris**. The estimated population of London is about **9.9 million** people, while the metropolitan population of Paris is approximately **13.2 million**

Agentic AI: Thought-Action-Observation cycle.



Source: Russel & Norvig: Artificial Intentiigence A modern approach (1995)

Agentic AI: Thought-Action-Observation cycle.



① THOUGHT

The model reasons about the question: what do I know, what do I need, and which tool or action should I use next?

② ACTION

The model calls a tool — e.g. a web search, API call, or code execution — with the parameters chosen in the Thought step.

③ OBSERVATION

The tool result returns. The model reads it and feeds it into the next Thought, looping until it can produce a final Answer.

↺ The loop repeats until the model decides it has enough information to give a final answer.

Source: Yao et al. (2022). ReAct: Synergizing Reasoning and Acting in Language Models. arXiv:2210.03629

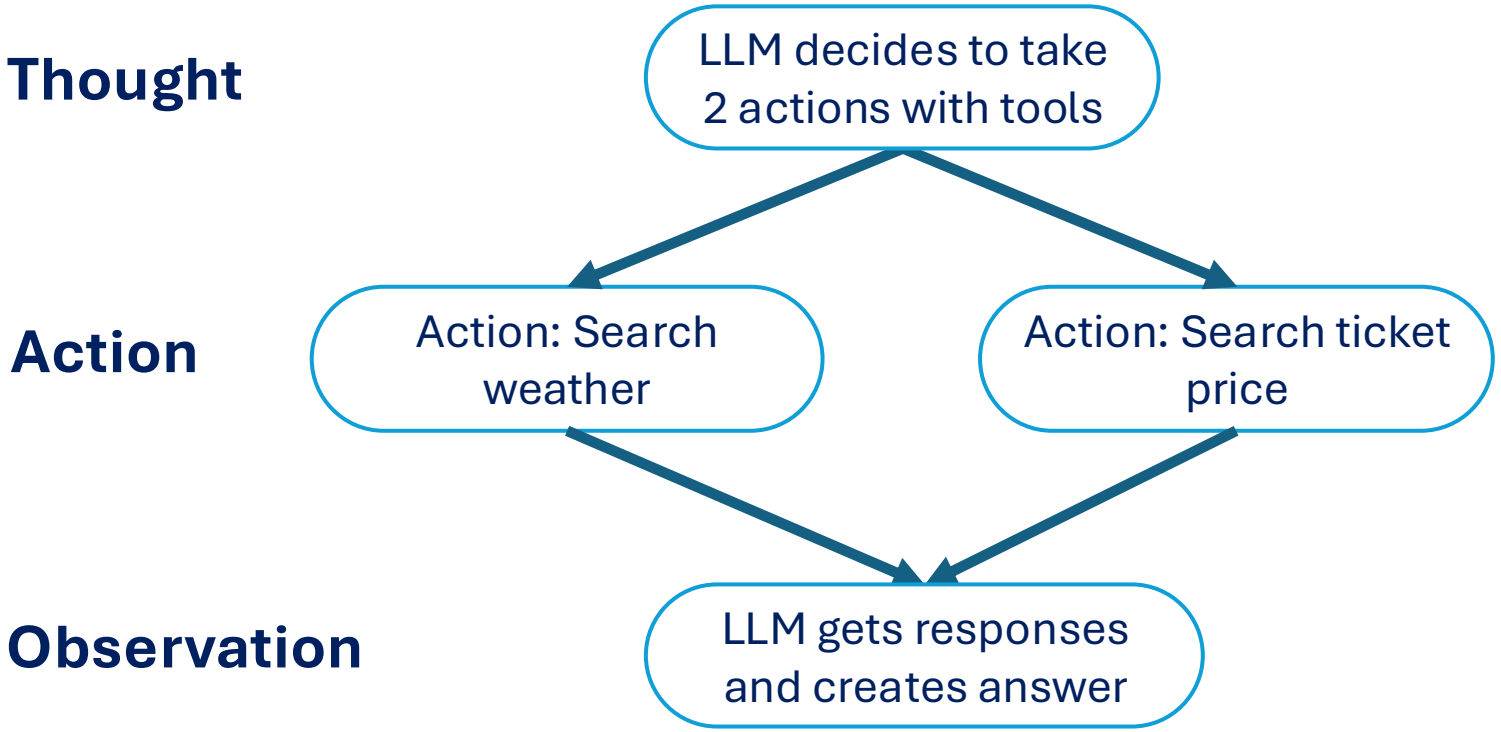
An AI Agent is an LLM that uses tool(s)

=

+

Agent LLM tool(s)

“I would like to go this weekend to Paris, but only if there is no rain and the train ticket from Amsterdam to Paris is cheaper than 100 euro.”



“The train ticket from Amsterdam to Paris is indeed cheaper than 100 euros. However, there is a possibility of rain in Paris this weekend.”

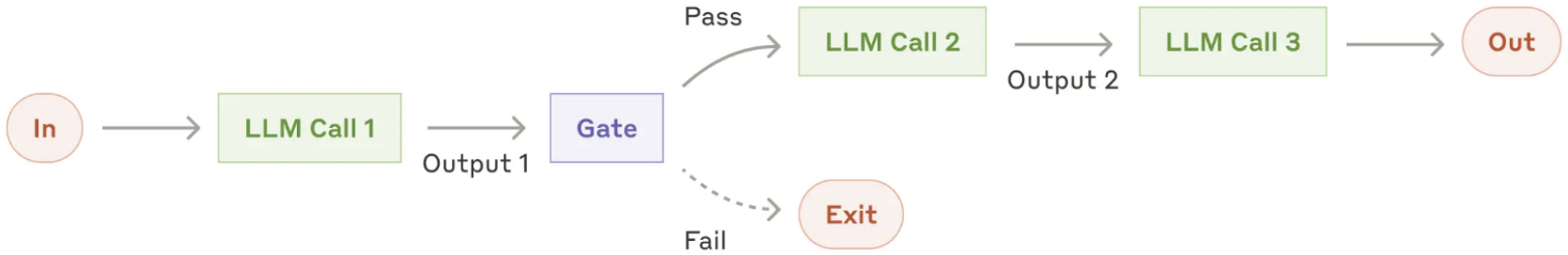
Assignment: trying out your first agent workflow

sense
maker
samst
erdam

Yes, I want to see some code:

No, I prefer no code:
https://huggingface.co/spaces/smolagents/computer-use-agent

From one agentic loop to multiple loops



The prompt chaining workflow

https://www.anthropic.com/engineering/building-effective-agents

Agentic frameworks

Agentic Frameworks

sense
maker
samst
erdam

Show

Smolagents

Wolfram alpha

Python

LangGraph
LlamaIndex

MS Autogen
Google ADK

Configuration

CrewAI
MetaGPT
N8N
OpenClaw

Many agentic frameworks exist, make sure you'll pick one that suits you!

Three frameworks compared

Framework	Ease to Learn	GitHub Stars	Popularity/Use Case
smolagents	Easy (code-first)	~26,300	Lightweight, research, fast prototyping
LangGraph	Moderate (graph-based)	~24,000	Enterprise, stateful workflows
LlamaIndex	Moderate (data-first)	~30,000	Data ingestion, RAG, search & retrieval

Exercise: compare the three frameworks

sense
maker
samst
erdam

Use the notebook / or colab: `Three_agentic_frameworks.ipynb`

This notebook requires ollama – please see next slide for more information.

run ollama

- Download and install via www.ollama.com (Mac / Windows / Linux)

- Start from de terminal (CLI):

  `ollama run tinyllama` (or any other model from ollama.com)
- Start chatting with the model!

- End ollama with CTRL+C or /bye or /exit
- You can then start another model.

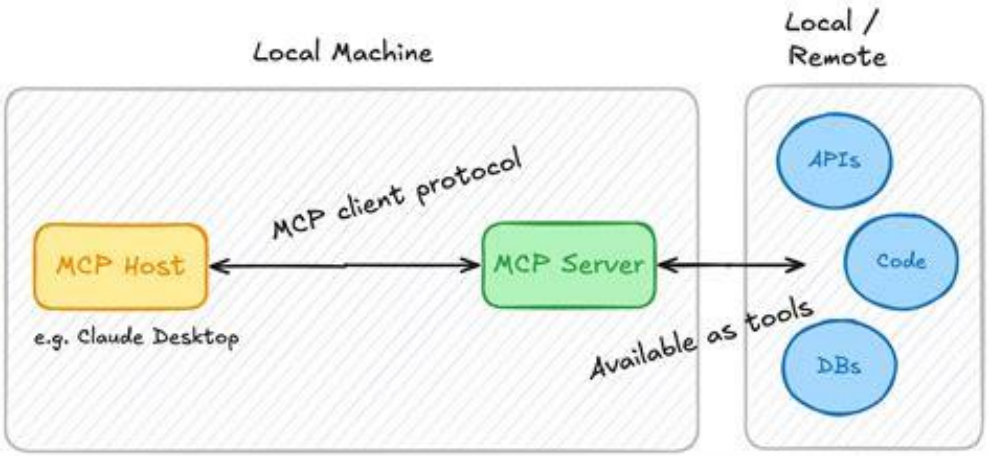
Two new standards for agents: MCP and Skills

What is MCP

- MCP (Model Context Protocol) is an open-source standard for connecting AI agents with external systems.

- Using MCP AI agents can connect to data sources, tools and workflows.

- Kind of like a specialized API language for agents!



MCP vs REST

sense
maker
samst
erdam

REST is the right choice when:

•You're building a service other developers or apps will consume
•You want standard HTTP tooling (caching, auth, rate limiting, load balancing)
•The consumer just needs data, not reasoning

MCP is best when:

- An LLM needs to decide *which* tool to call based on a user's natural language request
- You want one agent to expose capabilities to another agent
- You're building something where the set of available tools isn't known at code-write time

Give better instructions with Skills

sense maker samst erdam

Skills.MD is an open standard by Anthropic

It let’s you load a file that instructs the LLM what to do.

Most often used with Claude.
But it’s open source so you can run it with any LLM.

```
---
name: my-pptx-style
description: >
  Use this skill whenever creating or editing PowerPoint
  (.pptx) slides.
  Encodes the user's personal style: Aptos font, dark
  navy blue text on a
  white background. Trigger on any request to make,
  update, or design slides.
---

# My PowerPoint Style Guide

## Design System

### Colours
| Role | Hex | Usage |
|---|---|---|
| Primary text | `0D2D6B` | Titles, body, labels |
| Secondary text | `1565A7` | Descriptions, subtitles |
| Muted text | `4A6FA5` | Captions, source lines, hints |
```

Find skills at https://officialskills.sh/

MCP vs Agent Skills

Two complementary Anthropic open standards — they solve different problems and work best together

	Model Context Protocol (MCP)	Agent Skills (SKILL.md)
What it is	An open protocol (JSON-RPC over HTTP) that standardises how agents connect to external tools, APIs, databases, and services at runtime.	An open file format — a folder containing a SKILL.md plus scripts — that packages domain expertise and procedures an agent loads on demand.
Analogy	A universal plug socket. Any agent that speaks MCP can connect to any MCP server, regardless of who built either.	An onboarding guide on the shelf. The agent reads it only when the task matches, then acts on the instructions inside.
Strengths	• Live data & actions (calendar, DB, APIs) • Language/model agnostic • Standardised auth & discovery • Donated to Linux Foundation • 1000s of servers already exist	• Zero runtime overhead when not needed • Works with any model (Gemma, Llama…) • Version-controlled, shareable via Git • Progressive disclosure saves context tokens • Can bundle executable scripts
When to use	When the agent needs to take real-world actions or fetch live data: book a meeting, query a database, call a REST API, search the web.	When the agent needs to know how to do something: follow a company's report template, apply a classification ontology, run a specific workflow.
Open source?	Yes — spec donated to the Linux Foundation (Dec 2025). Implemented by Claude, OpenAI, Gemini CLI, Cursor and many others.	Yes — open standard published Dec 2025. Compatible with Claude Code, Gemini CLI, Codex CLI, Cursor, OpenCode and more.

Used together: MCP gives the agent access to tools & live data. Skills teach it how to use them correctly in your specific context.

Sources: Anthropic (2024). Model Context Protocol — modelcontextprotocol.io · Anthropic (2025). Agent Skills — platform.claude.com/docs/agents-and-tools/agent-skills

Exercise get started with MCP

sense
maker
samst
erdam

Use the Jupyter Notebook:

`FastMCP.ipynb`

Danger Zone:
towards autonomous agents

You can give your agent more autonomy

sense maker samst erdam

Would you:

- Let an agent do a search?
- Give an agent access to write your mails?
- Give an agent access to your file system?

OpenClaw

Tech Industry > Artificial Intelligence

AI tool OpenClaw wipes the inbox of Meta's AI Alignment director despite repeated commands to stop — executive had to manually terminate the AI to stop the bot from continuing to erase data

News By Bruno Ferreira published February 24, 2026

It's almost like we could all see this one coming.

Also: OpenClaw was released by an Independent Developer. Not by a big lab. The big labs saw the risks and did not want to release this. Classic Innovators Dilemma.

sense
maker
samst
erdam

Autonomous agents are a risk.
So give them guardrails.

Example guardrails

- Principle of least privilege / Role based access management
- Reversible versus irreversible actions.
- Logging with alerts
- Four-eyes principle

These principles are not new, we haven't applied them to AI Agents yet.

From Tool Caller to Autonomous Agent

Five levels of AI agent autonomy · Feng, McDonald & Zhang (2025)

L1 Operator

Human drives everything

User controls every action. Agent only responds when explicitly asked. No autonomous steps.

EXAMPLE

Single tool call, one result, done

L2 Collaborator

Agent suggests, human decides

Agent proposes next steps and drafts actions. Human approves each before execution.

EXAMPLE

Agent proposes train + weather check; user confirms before each call

L3 Consultant

Agent acts, human monitors

Agent executes multi-step plans autonomously. Human sets goal and reviews outcome — not each step.

EXAMPLE

Paris weekend planner runs both tools, reasons, returns verdict

L4 Approver

Agent decides, human vetoes

Agent operates continuously and handles exceptions. Human only intervenes to block or redirect.

EXAMPLE

Multi-agent system: orchestrator delegates to specialist sub-agents

L5 Observer

Agent acts fully autonomously

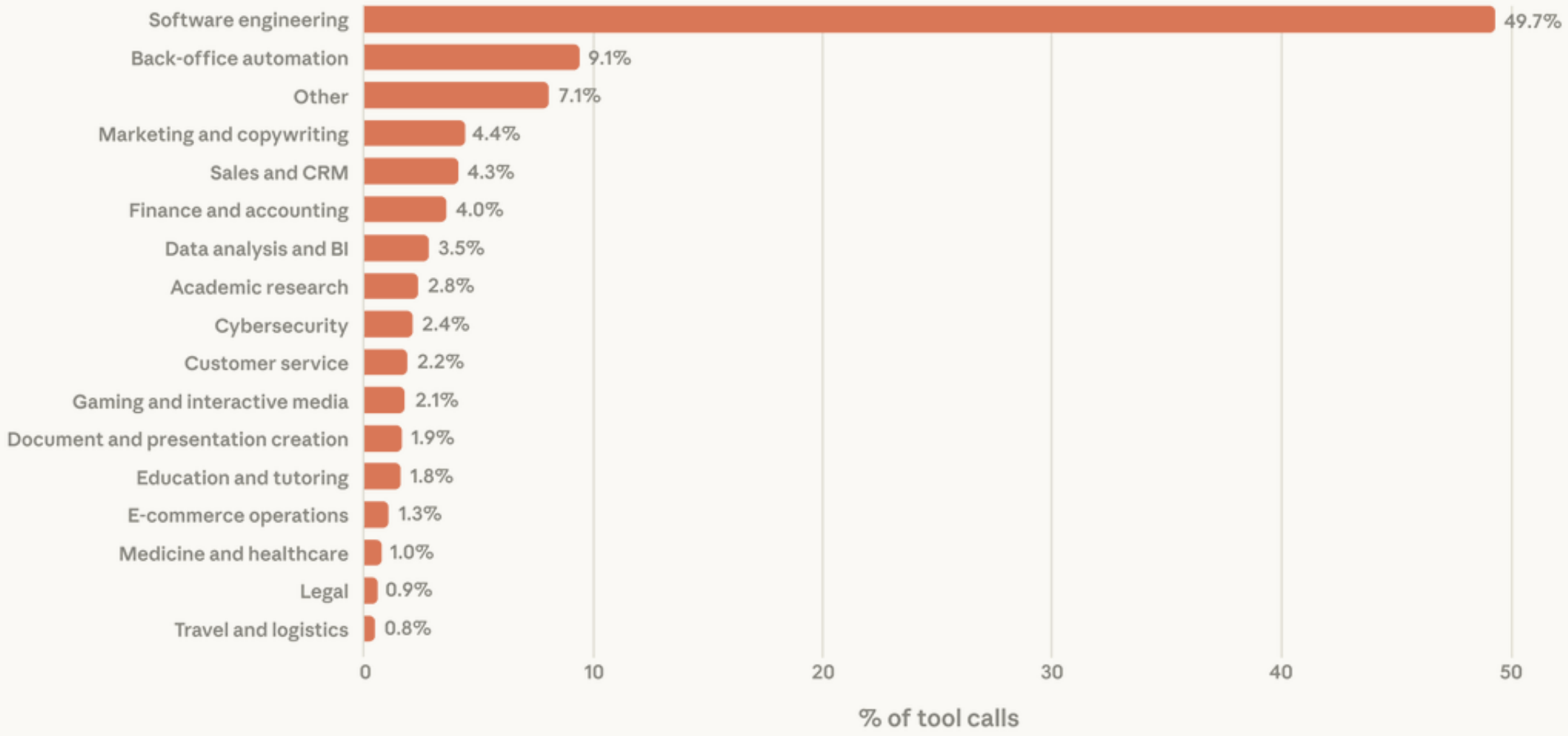
Human defines policy upfront. Agent monitors, acts, and reports. No per-task involvement.

EXAMPLE

Autonomous sound classifier that detects, logs and escalates city noise events

Source: Feng, McDonald & Zhang (2025). Levels of Autonomy for AI Agents. arXiv:2506.12469 / Knight First Amendment Institute, Columbia University. https://arxiv.org/pdf/2506.12469

In what domains are agents deployed?



Multi-agent systems

Example: city noise report pipeline — three agents, one task

Task: *"Measure background noise at 5 Amsterdam intersections and produce an ISO 9614 compliance report."*

① Planner

Generates high-level tasks

RESPONSIBILITIES

- Decompose goal into subtasks
- Assign locations & priorities
- Set success criteria per task
- Sequence tasks & set timing

OUTPUT

Ordered task list with criteria

tasks →

② Worker

Executes the tasks

RESPONSIBILITIES

- Call measurement API per location
- Apply background subtraction formula
- Retrieve city noise ontology via MCP
- Write sections of the ISO report

OUTPUT

Measurements + draft report

output →

③ Verifier

Controls every result

RESPONSIBILITIES

- Check margin ≥ 3 dB per measurement
- Validate ISO 9614 format compliance
- Flag uncertain results for re-run
- Approve or reject each section

OUTPUT

Verified report or rejection + reason

← reject / replan

Pattern: Generator–Verifier (P1) nested in Orchestrator–Subagent (P2) · Verifier = vier-ogen principe — no result passes without independent review

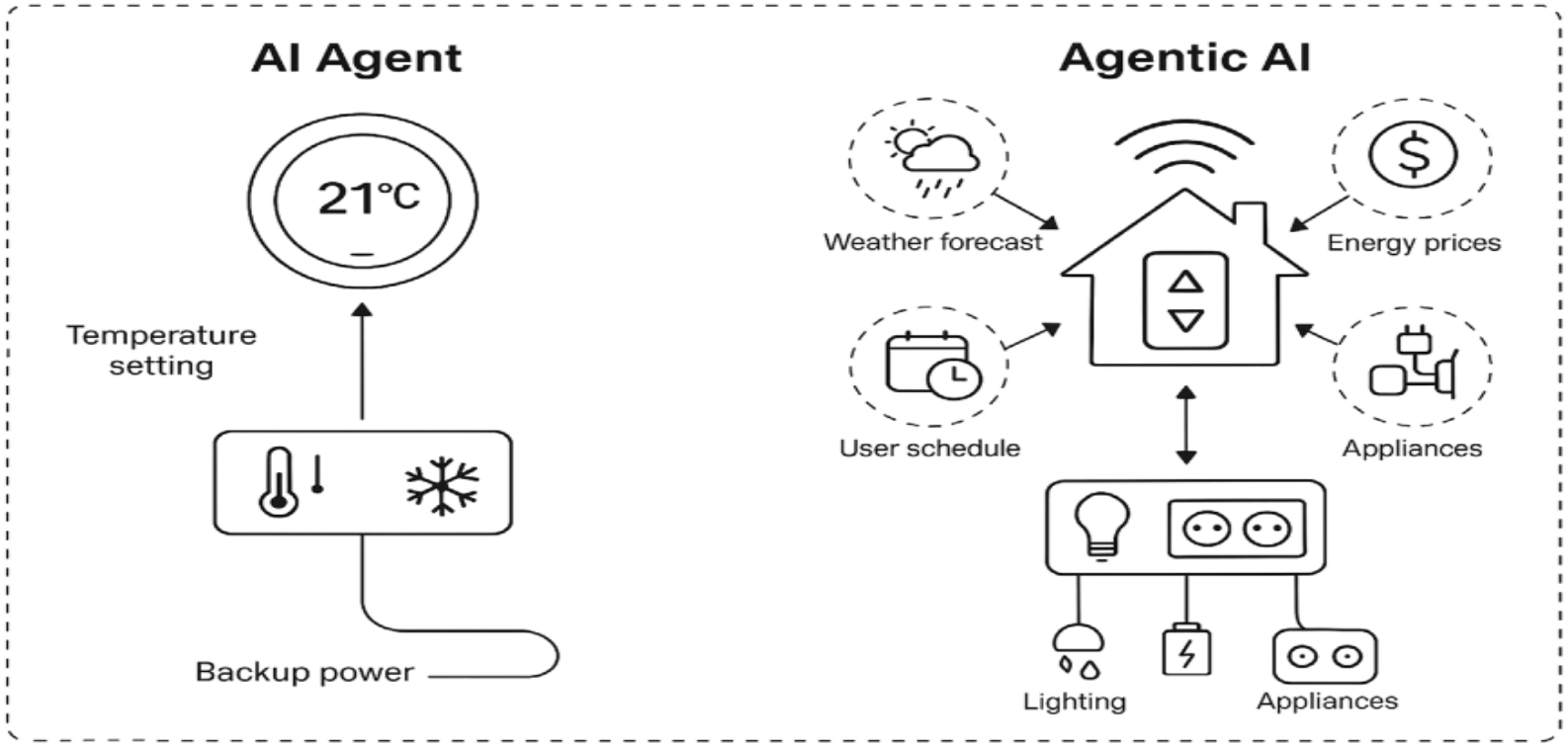
Roles: Planner (orchestrator) · Worker (subagent) · Verifier (judge) · Ref: Feng, McDonald & Zhang (2025). Levels of Autonomy for AI Agents. arXiv:2506.12469

Our future…?

© marketoonist.com

Annexes

AI Agent vs Agentic AI



Levels of autonomy for coding agents

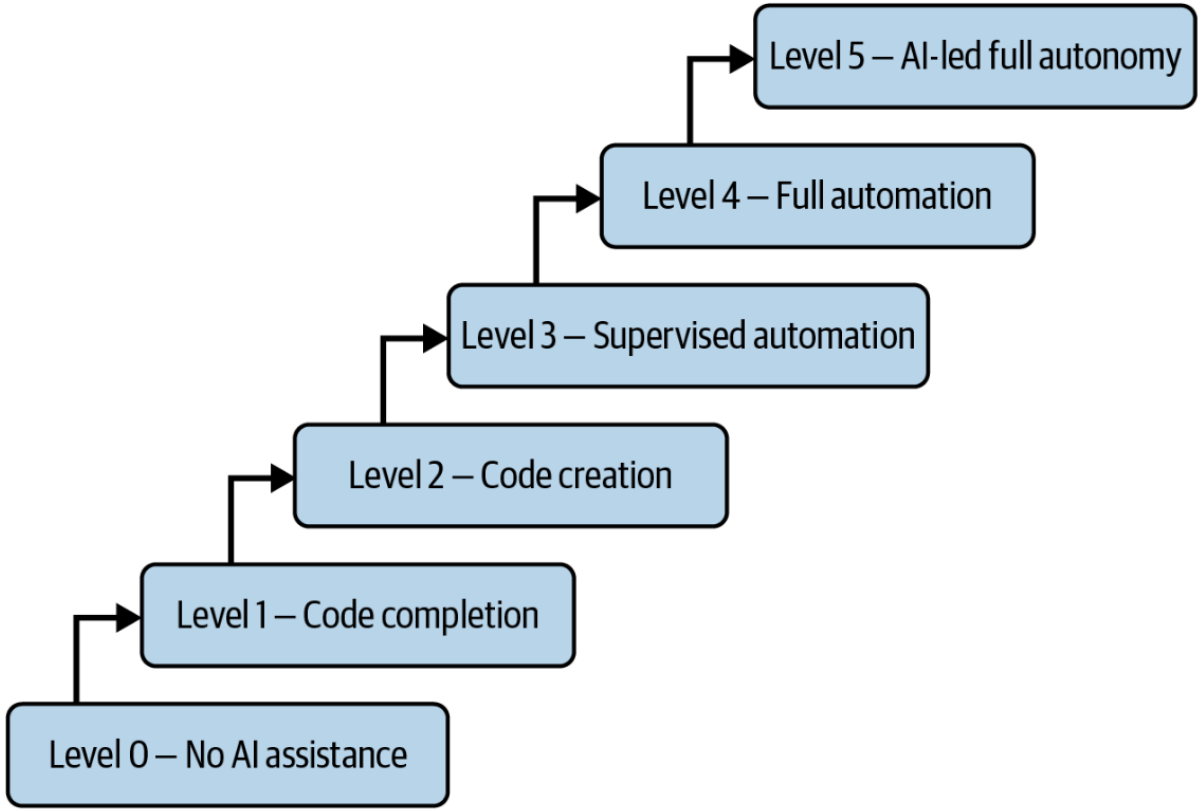
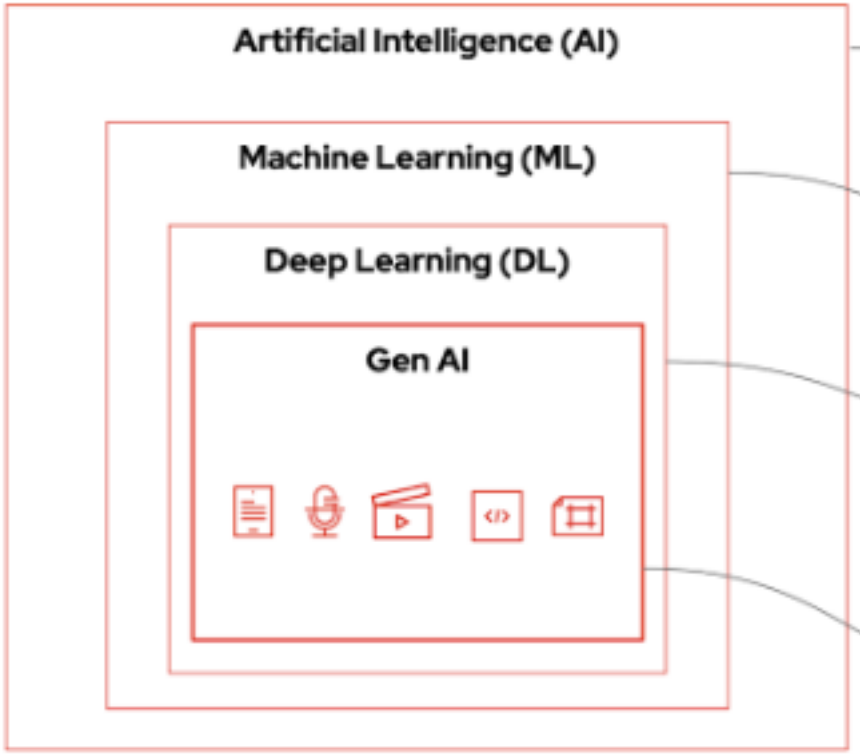


Figure 2-1. Programming systems have different levels of AI capability

GenAI in the context of AI



Artificial Intelligence

A multidisciplinary field of Computer Science that aims to create systems capable of emulating and surpassing human-level intelligence

Machine Learning

A subfield for training computers to learn and make decisions from data, without explicit programming

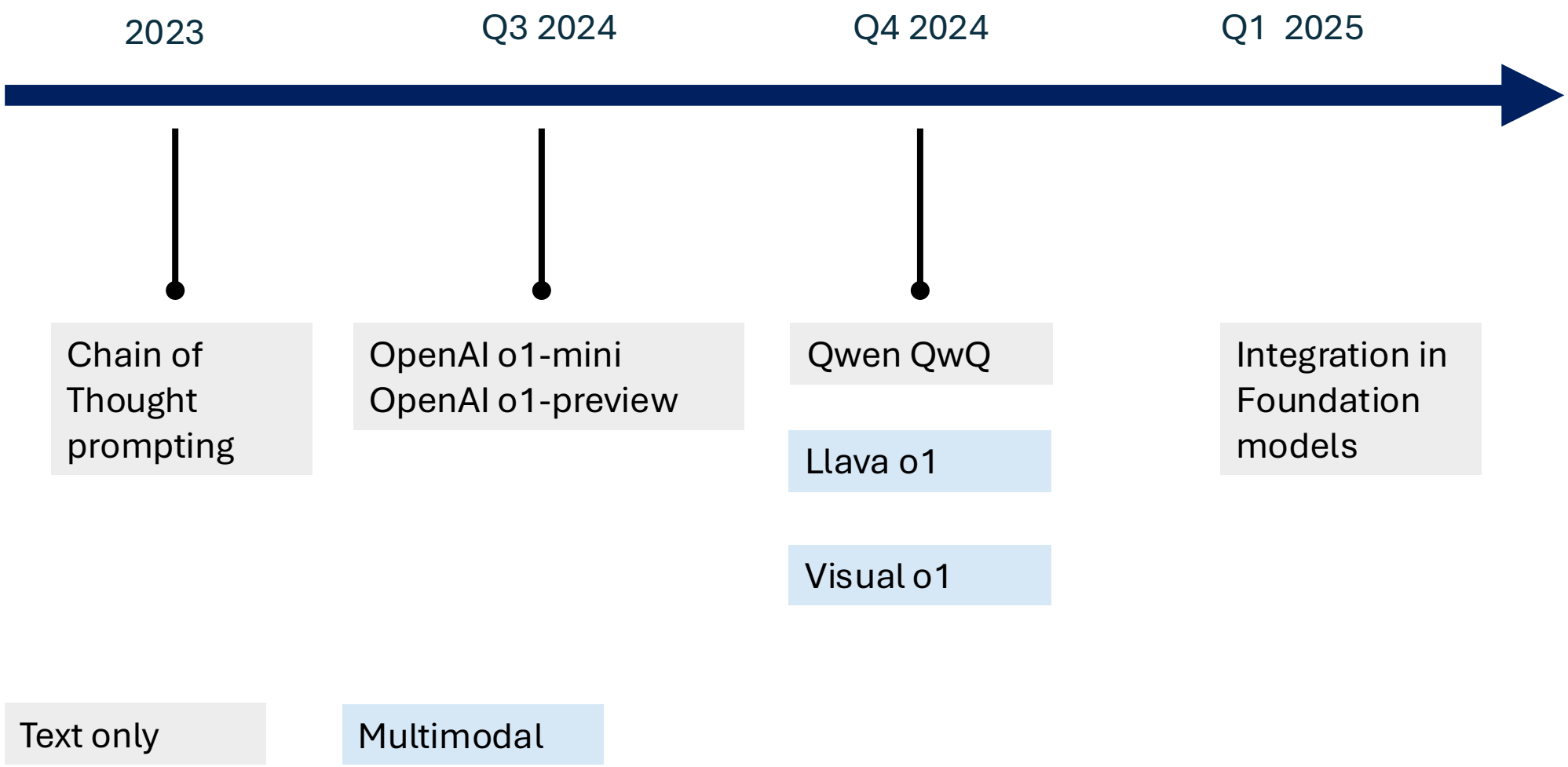
Deep Learning

A subset from ML involved neural networks enabling complex pattern recognition and analysis from data

Gen AI Learning

Subfield focusing on generating new content such as text, images, code and more

Timeline reasoning models



Coding agent vs coding harness

- A **coding agent** (Claude Code, Devin, Cursor) is an opinionated *product* that makes decisions about how you work — it has built-in plan modes, permission flows, sub-agents, to-do management. You adapt your workflow to it.

- A **coding harness** is the minimal *connective structure* that holds the pieces together — the model, the terminal UI, the context window, the session history — and then gets out of the way. You decide what goes in it.

Agent frameworks compared

Framework	Focus/Strengths	GitHub Stars	Best For
AutoGen	Conversational agents, research	~26,000	Research, complex conversations
Semantic Kernel	Enterprise, .NET integration	~27,000	.NET teams, Azure users
OpenAI Agents SDK	Native OpenAI integration	N/A	OpenAI-centric projects
Google ADK	Google Cloud, Vertex AI	~17,000	Google Cloud users
MetaGPT	Role-based, organizational structure	N/A	Structured, role-based workflows
OpenClaw	Configuration-first	N/A	Declarative agent orchestration
CrewAI	Easy (role-based)	~45,900–48,000	Multi-agent teams, rapid iteration